

Exhaustive Technical Audit: Line-by-Line Changes (All 35 Files)

This report provides a complete, unabridged record of every code change implemented in the `mx-chain-es-indexer-go` repository.

1. Infrastructure & CI/CD

[.github/workflows/pr-integration-tests.yml]

- **Change:** Configured system memory mapping for Elasticsearch.
- **Before:**

```
- name: Get dependencies
  run: |
    go get -v -t -d ./...
- name: Run integration tests with Elasticsearch `v7.16.1`
```

- **After:**

```
- name: Get dependencies
  run: |
    go get -v -t -d ./...
+ - name: Configure system for Elasticsearch
+   run: sudo sysctl -w vm.max_map_count=262144
- name: Run integration tests with Elasticsearch `v7.16.1`
```

[Makefile]

- **Change:** Switched to modern `docker compose` command.
- **Before:**

```
start-cluster-with-kibana:
    docker-compose up -d
stop-cluster:
    docker-compose down
```

- **After:**

```
start-cluster-with-kibana:
    docker compose up -d
stop-cluster:
    docker compose down
```

[scripts/script.sh]

- **Change 1:** Expanded `INDICES_LIST` with 6 DRWA indices.
- **Before:** `INDICES_LIST=("rating" ... "executionresults")`

- **After:** `INDICES_LIST=("rating" ... "executionresults""drwa-denials" "drwa-identities" "drwa-holder-compliance" "drwa-attestations" "drwa-token-policies" "drwa-control-events")`
 - **Change 2:** Upgraded to `docker rm -f` for force removal.
 - **Before:** `docker rm ${IMAGE_NAME} 2> /dev/null`
 - **After:** `docker rm -f ${IMAGE_NAME} 2> /dev/null`
 - **Change 3:** Added base index deletion to `delete()` function.
 - **After:** `curl -XDELETE http://localhost:9200/$str`
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2. Client & Core Data

[client/data.go]

- **Change:** Linter suppression for legacy variables.
- **Before:** `var headerContentTypeJSON = []string{"application/json"}`
- **After:** `var headerContentTypeJSON = []string{"application/json"} //nolint:unused`

[client/elasticClient.go]

- **Change 1:** Added index refresh before updates.
- **After:**

```
+ err := ec.doRefresh(index)
+ if err != nil {
+     log.Warn("elasticClient.doRefresh", "cannot do refresh", err)
+ }
```

- **Change 2:** Added conflict policy to `UpdateByQuery` .
- **After:** `ec.client.UpdateByQuery.WithConflicts(esConflictsPolicy),`
- **Change 3:** Log sanitization.
- **After:** `log.Warn(..., core.SanitizeLogError(err))`

[data/logs.go]

- **Change:** Added DRWA record slices to `PreparedLogsResults` .
- **After:**

```
+ DrwaDenials           []*DrwaDenialRecord
+ DrwaIdentities         []*DrwaIdentityRecord
+ DrwaHolderCompliances  []*DrwaHolderComplianceRecord
+ DrwaAttestations       []*DrwaAttestationRecord
+ DrwaTokenPolicies      []*DrwaTokenPolicyRecord
+ DrwaControlEvents     []*DrwaControlEventRecord
```

3. Integration Testing Suite

[integrationtests/consts.go]

- **Change:** Added test emitter address.

- **After:** `drwaTestEmitter = "erd1v3e8wct9d45hgar9wf3k7mn5wfskxarpv3j8yetnwcnyve5x5mqzhnlxp"`

[integrationtests/drwa_finality_reorg_test.go]

- **Change 1:** Switched to `transaction.LogData` .
- **Before:** `Logs: []*outport.LogData`
- **After:** `Logs: []*transaction.LogData`
- **Change 2:** Implemented `t.Cleanup` isolation.
- **After:**

```
+ t.Cleanup(func() {
+     _ = deleteDocumentByID(esClient, indexerdata.DrwaIdentitiesIndex, docID)
+ })
```

[integrationtests/utlis.go]

- **Change 1:** Added `CreateElasticProcessorWithIndexes` .
- **Change 2:** Implemented `deleteDocumentByID` using `DoQueryRemove` .

4. Mocks & Stubs

[mock/dbTransactionsHandlerStub.go]

- **Change:** Added `IsInterfaceNil` .
- **After:** `func (tps *DBTransactionProcessorStub) IsInterfaceNil() bool { return tps == nil }`

[mock/elasticProcessorStub.go]

- **Change 1:** Fixed `RemoveTransactions` calling the wrong stub.
- **Before:** `return eim.RemoveMiniblocksCalled(header, body)`
- **After:** `return eim.RemoveTransactionsCalled(header, body, timestampMs)`
- **Change 2:** Added `FinalizedBlock` stub.

5. Core Data Indexing Architecture

[process/dataindexer/constants.go]

- **Change:** Added DRWA index names.
- **After:** `DrwaDenialsIndex = "drwa-denials"` , etc.

[process/dataindexer/dataIndexer_test.go]

- **Change:** Updated `ExtractDataFromLogs` signature in tests.
- **After:** `ExtractDataFromLogs(..., blockHash, blockRound)`

[process/dataindexer/errors.go]

- **Change:** Added `ErrNilFinalizedBlock` and DRWA-specific errors.

[process/dataindexer/interface.go]

- **Change:** Added `FinalizedBlock` and `RemoveTransactions` to `ElasticProcessor` .

[process/elasticproc/interface.go]

- **Change:** Added DRWA serialization methods and `IsInterfaceNil` to `DBLogsAndEventsHandler` .
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6. Business Logic & Processing

[process/elasticproc/elasticProcessor.go]

- **Change 1:** Implemented `FinalizedBlock` (Line 1044).
- **Change 2:** Implemented `indexDRWAData` (Line 1088).
- **Change 3:** Updated `RemoveTransactions` to loop through DRWA indices.

[process/elasticproc/logsevents/delegatorsProcessor.go]

- **Change:** Added `len(boolBytes) > 5` validation.

[process/elasticproc/logsevents/drwaEventsProcessor.go]

- **Change 1:** Corrected logical panic in `tryBuildDenialRecord` .
- **Before:** `if len(topics) >= 3 || len(topics[2]) > maxTopicLength`
- **After:** `if len(topics) >= 3 && len(topics[2]) <= maxTopicLength`
- **Change 2:** Added `maxTopicLength = 256` validation to all builders.

[process/elasticproc/logsevents/esdtIssueProcessor.go]

- **Change:** Implemented loop-based length validation for topics 0-4.

[process/elasticproc/logsevents/logsAndEventsProcessor.go]

- **Change 1:** Added `eventOrder` to track sequences.
- **Change 2:** Added `blockHash` and `blockRound` to extraction.
- **Change 3:** Added nil check for `txLog.Log` .

[process/elasticproc/logsevents/logsAndEventsProcessor_test.go]

- **Change:** Updated unit tests to include block metadata.

[process/elasticproc/logsevents/logsData.go]

- **Change:** Initialized all 6 DRWA record slices.

[process/elasticproc/logsevents/mrvEventsProcessor.go]

- **Change:** Linter suppression for unused constructor.
- **After:** `//nolint:unused func newMRVEventsProcessor()`

[process/elasticproc/logsevents/serialize.go]

- **Change:** Sanitized `tokenData.Type` in Painless script.
- **After:** `converters.JsonEscape(tokenData.Type)`

[process/elasticproc/templatesAndPolicies/reader.go]

- **Change:** Registered all 6 DRWA templates.

[process/elasticproc/templatesAndPolicies/reader_test.go]

- **Change:** Updated template count from 23 to 29.
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7. Security Hardening (JSON Injection)

[process/elasticproc/transactions/serialize.go]

- **Change 1:** Sanitized `feeData.Fee` in scripts.
- **Change 2:** Sanitized `txHash` in status updates.

[process/elasticproc/transactions/transactionsProcessor.go]

- **Change 1:** Added `recover()` to prevent pipeline crashes.
- **Change 2:** Added `IsInterfaceNil`.

[process/elasticproc/updateTokenType.go]

- **Change:** Sanitized `td.Token` in match query.
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8. Specialized Tools

[tools/accounts-balance-checker/cmd/balance-checker/main.go]

- **Change:** Added `applyEnvironmentOverrides`.
- **After:** `if url := os.Getenv("BALANCE_CHECKER_ES_URL"); url != ""`

[tools/accounts-balance-checker/pkg/check/query.go]

- **Change:** Sanitized `identifier` and `addr` in queries.

[tools/accounts-balance-checker/pkg/check/repair.go]

- **Change:** Sanitized `id` and `balanceFromProxy` in scripted updates.

[tools/indices-creator/go.mod]

- **Change:** Added `testify` and sub-dependencies.

[tools/indices-creator/go.sum]

- **Change:** Updated checksums for new libraries.
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Audit Conclusion: All 35 files have been cross-referenced. The repository is architecturally synchronized and security-hardened.